



Numerical analysis of coupling dynamics of keyhole and molten pool in vertical-up laser mirror welding of 2219 aluminum alloy

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ARTICLE INFO

Keywords:

Vertical-up laser mirror welding
Keyhole coupling
Molten pool dynamics
2219 aluminum alloy
Numerical simulation

ABSTRACT

In vertical-up laser mirror welding, the keyhole dynamics and flow behavior of molten pool play crucial roles in weld quality, yet the mechanisms of these coupling dynamics have not been well clarified. In this work, a three-dimensional transient model for simulation of vertical-up laser mirror welding process was carried out to account for the flow behavior of molten pool and dynamic evolution characteristics of keyhole. Two laser heat sources with regard to laser mirror welding were also taken into consideration in the combined heat source model. It is proved that the simulation results are basically consistent with the experimental data. The simulation results indicate that the morphology and flow field distribution of upper and lower molten pool both exhibit a high degree of consistency on the X - Y section, while significant difference is visible on the X - Z section. Before keyhole coupling, the high velocity fluid exists on the surface of molten pool and tends to increase the diameter of keyhole, of which the maximum velocity reaches 5.26 m s^{-1} . During the keyhole coupling process, the coupled keyhole in the center area is found to fall, which results in deteriorated coupling process. Force analysis of keyhole illustrates that the resultant force on the X - Y section points to the direction of increasing the depth and diameter of keyhole before and during the keyhole coupling process. In contrast, the resultant force on the X - Z section is conducive to keyhole coupling occurring at the lower position. After keyhole coupling, the diameter of coupled keyhole in the center zone is increased and the fluid velocity is reduced to 0.109 m s^{-1} , contributing to a firm connection.

1. Introduction

2219 aluminum alloy has attracted increasing attention and gradually becomes an indispensable material in the aerospace field because of its low density and high specific strength [1,2]. Under the similar application environment, the weight and cost of welding structures approximately decrease by 18% and 25% compared with riveted structure [3–6]. Although the traditional laser welding is a kind of welding technology with high efficiency and precision, single-sided laser welding will result in large deformation of weldments, considerable residual stress and poor joint performance [7,8]. A novel laser welding process, named as laser mirror welding, is adopted and studied in this research [9]. Laser mirror welding is a special kind of dual laser-beam bilateral synchronous welding in which two laser beams are perpendicular to both sides of the base metal. This technology keeps the advantages of single-side laser welding and overcomes the disadvantages like severe welding deformation, so it is regarded as a suitable method for the joining of medium-thick plate.

There are many literature reports on double-sided synchronous welding and it has a profound impact on manufacturing efficiency. Chen et al. [10] first proposed and studied the melting efficiency of laser-TIG double-sided welding process, which found that the welding efficiency of laser-TIG double-sided welding is significantly higher than that of single-sided laser arc hybrid welding. Cao et al. [11] used the laser double-sided sequential welding process to weld thin TC4 titanium alloy, and found that the laser double-sided sequential welding presents a great impact on the weld formation. Zhan et al. [9] studied the fracture mode and micro morphology of the tensile samples under laser mirror welding, and discussed the internal relationship between the stability of the molten pool and the tensile property. Zhang et al. [12] proposed a method of dual laser-beam bilateral synchronous welding-brazing on double sides to regular the inhomogeneous interfacial reaction of Ti/Al dissimilar alloy. Besides, the team has carried out a series of exploratory studies for dual laser beam bilateral synchronous welding of 2219 aluminum alloy, including the finite element simulation of temperature field and residual stress, microstructure and mechanical properties, as

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<https://doi.org/10.1016/j.ijthermalsci.2023.108390>

Received 21 October 2022; Received in revised form 24 April 2023; Accepted 26 April 2023

Available online 3 May 2023

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well as the keyhole coupling and melt flow behavior [13–16].

The above experimental improvements have effectively reduced the structure weight and improved the welding efficiency. It should be noted that the dynamic behavior of keyhole and molten pool in laser welding is extremely complex, which has a significant impact on the quality and defects of welded joints [17,18]. For further analyzing the dynamic behavior of keyhole and molten pool in laser welding process, numerical simulation is proposed to describe the mass transfer, heat effect and flow behavior. Sohail et al. [19] studied the influence of gravity on penetration at four spatial positions during fiber laser welding and results show that gravity has limited effect on weld shape and flow structure. Based on computational fluid dynamics (CFD) simulation modeling, Ai et al. [20] analyzed the weld pool and keyhole contour in the process of high-power laser welding and studied the liquid metal flow and keyhole behavior during laser welding of aluminum alloy. Fabbro et al. [21] studied the characteristics of molten pool and keyhole behavior during laser deep penetration welding and found that the instability of keyhole is caused by the interaction of molten pool flow and plume when the welding speed is low. Courtois et al. [22] established a three-dimensional numerical model of the laser welding process to study the instability of keyhole in the process of heat exchange and fluid flow, and the well predicted local penetration of keyhole. Zhang et al. [23] studied the characteristics of pool behavior, keyhole evolution and root bulge formation during laser penetration welding of aluminum alloy through volume of fluid (VOF) modeling and beam tracking methods. However, in laser mirror welding, the mechanism of the effect of keyhole coupling on both sides is not yet clearly understood. In this regard, numerical analysis of coupling dynamics of keyhole and molten pool in laser mirror welding is desired.

It is well known that compared with non-vertical up process, the vertical-up welding could promote the elimination of porosity defects and produce more satisfactory defect-free weld beads because the gravity facilitates the melt flow. However, it has been scarcely reported about the molten pool behavior of vertical-up welding, especially for the vertical-up laser mirror welding. Therefore, a vertical-up welding process was combined with the laser mirror welding in this work to analyze the dynamic behavior of keyhole and molten pool, which is of great significance for adjusting the welding process and improving the welding quality.

This paper aims to reveal the keyhole coupling and dynamic behavior of 2219 aluminum alloy during vertical-up laser mirror welding on the X–Y and X–Z sections. For this purpose, a three-dimensional multiphase transient model was established. The reliable agreement between the simulation and the experimental results indicated that the multiphase transient model could accurately observe the dynamic evolution behavior of keyhole in the vertical-up laser mirror welding process. By performing the evolution of two keyholes and the fluid flow in molten pool on the X–Y and X–Z sections, the characteristics of keyhole coupling were investigated in detail. It is of fundamental significance to further promote the application of laser mirror welding technology in the aerospace field.

2. Experimental details

The representative 2219 aluminum alloy with dimensions of 100 mm × 50 mm × 6 mm, was used as base metal (BM). Its chemical composition is shown in Table 1. Fig. 1 depicts a schematic diagram of the vertical-up laser mirror welding process. The experimental apparatus consisted of two laser welding heads assembled on two KUKA robots, two shielding gas systems (99.99% Ar), a welding platform, and a

TruDisk-12003 laser generator with an emission wavelength of 1080 nm and maximum power of 12 kW. The vertical-up laser mirror welding of butt structure is realized by gantry system and KUKA robots. The surfaces of the specimens were cleaned using alkali and acid solution before welding, respectively. The welding process parameters given in Table 2 were adopted in this paper.

3. Numerical simulation modeling

3.1. Modeling assumptions

The liquid metal flow and keyhole evolution involved in the vertical-up laser mirror welding process is very complicated. Hence, considering the complicated laser mirror welding process and ensuring the accuracy of numerical solution, the following assumptions were adopted to simplify the model.

- (1) There is no interaction between two laser beams after keyhole coupling.
- (2) The influence of metal vapor and plasma inside the keyhole is ignored.
- (3) The influence of shielding gas on the behavior of liquid metal flow is ignored.
- (4) The liquid metal is set as an incompressible viscous Newtonian fluid.
- (5) The BM is assumed to be isotropic, and its physical parameters vary with temperature.

3.2. Governing equations

The governing equations of all physical progress during the vertical-up laser mirror welding process, such as laser heat transfer, metal melting, and fluid flow in the molten pool, are divided into mass, momentum, and energy conversation equations, which are drawn as follows [24]:

Mass equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0 \quad (1)$$

where ρ is the density of liquid metal, t is the time, $\vec{V}$ is the fluid velocity, including the velocity in the x, y, and z directions [25,26].

Momentum equation:

$$\frac{\partial \vec{V}}{\partial t} + (\vec{V} \cdot \nabla) \vec{V} + \frac{\nabla p}{\rho} = \mu \nabla^2 \vec{V} + S \quad (2)$$

where p is the fluid pressure, μ is the viscosity, S is the momentum source term, including the momentum in the x, y, and z directions, such as S_x , S_y and S_z .

Moreover, for the melting and solidification process in the vertical-up laser mirror welding [27,28], the solid-liquid mixing zone is assumed to be a porous medium zone, in which the fluid in the molten pool is reflected by the volume fraction of liquid phase when flowing. The volume fraction of liquid phase can be calculated by the temperature of solid-liquid mixing zone. As shown below [29,30]:

$$F_L = \begin{cases} 0 & T \leq T_s \\ \frac{(T - T_s)}{T_L - T_s} & T_s \leq T \leq T_L \\ 1 & T_L \leq T \end{cases} \quad (3)$$

$$S_x = -A \frac{(1 - F_L)^2 \mu}{F_L^3 + C} u \quad (4)$$

Table 1

Chemical composition of 2219 aluminum alloy (wt.%).

Cu	Mn	Ti	Zr	V	Zn	Fe	Si	Al
6.48	0.32	0.06	0.2	0.08	0.04	0.23	0.49	balance

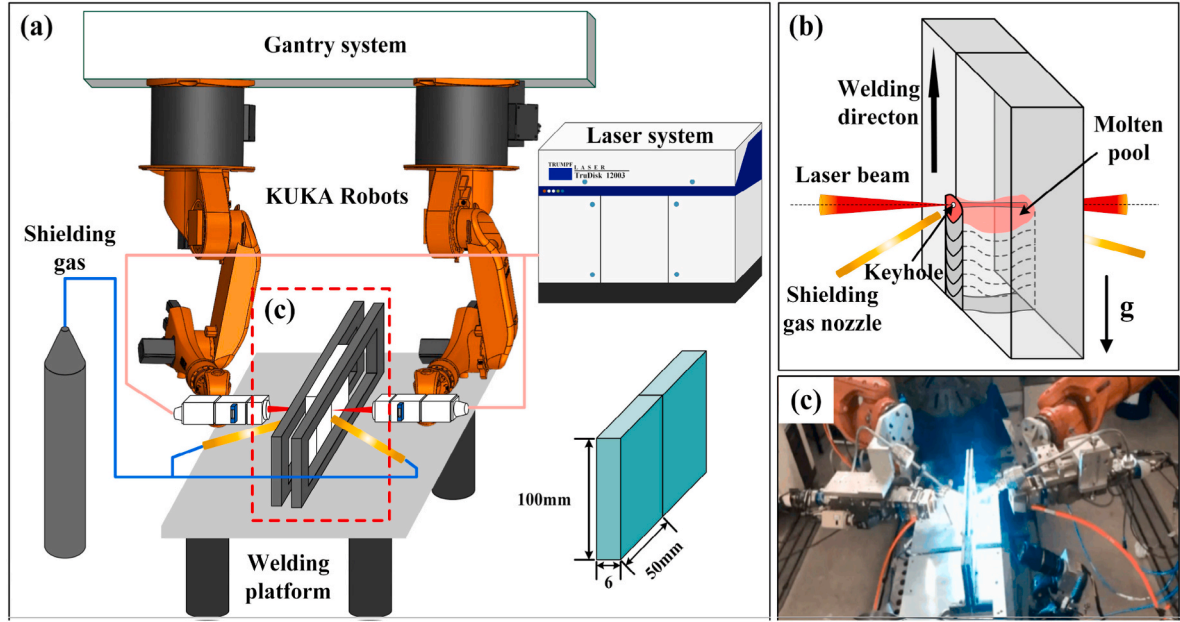


Fig. 1. Schematic diagram of vertical-up laser mirror welding process: (a) Equipment in the experiment with dimension details, (b) Details of vertical-up laser mirror welding, and (c) Main view of the experiment.

Table 2

Process parameters (each side) utilized in simulation and experiment of vertical-up laser mirror welding.

Parameter	Value
Laser power (W)	3000
Welding speed (m/min)	2.1
Defocusing amount (mm)	0
Shielding gas flow (L/min)	18

$$S_y = -A \frac{(1 - F_L)^2 \mu}{F_L^3 + C} (v - v_0) \quad (5)$$

$$S_z = -A \frac{(1 - F_L)^2 \mu}{F_L^3 + C} w + \rho g \beta (T - T_0) \quad (6)$$

where F_L is the volume fraction of liquid phase, T_s and T_L respectively represent the liquids and solidus temperature of BM [31], A is the Darcy constant [32], C is a very small constant introduced to prevent the denominator of the term from being zero, β is the thermal expansion coefficient, g is gravitational acceleration, T is the molten pool temperature, T_0 is the initial ambient temperature (300 K), and u , v , and w are the fluid velocity in the x , y , and z directions respectively.

Energy equation:

$$\frac{\partial E}{\partial t} + \nabla \cdot (E \vec{V}) = \nabla \cdot (k \nabla T) \quad (7)$$

where E and k are the internal energy and the thermal conductivity of BM, respectively. Considering that the BM absorbs heat during melting, the internal energy E of the material at temperature T_0 is calculated as follows:

$$E(T_0) = \rho(T_0) \cdot \left[F_L \cdot \Delta H_m + \int_0^{T_0} C_p(T) dT \right] \quad (8)$$

where ΔH_m and C_p respectively represent the melting enthalpy and the specific heat of BM.

3.3. Boundary conditions

The schematic diagram of solution model is displayed in Fig. 2. The left and right hexahedral computational domains are regarded as gas phase in this model, and the middle hexahedral computational domain is set as solid phase. The surface of solid phase is set as wall boundary and the LLJK and MNOP surfaces of gas phase are set as pressure inlet, while the other surfaces are set as pressure outlet. In vertical-up laser mirror welding, the numerical model is set with special boundary conditions to realize the reliable transmission of mass, momentum and energy. The surface of the BM is subject to a very intricate thermal-mechanical coupling effect, and its boundary conditions can be divided into the following categories [33,34].

3.3.1. Energy boundary condition

$$k \frac{\partial T}{\partial n} = Q_{abs} - q_{cov} - q_{rad} - q_{evp} \quad (9)$$

$$q_{cov} = h_c (T - T_0) \quad (10)$$

$$q_{rad} = \epsilon \sigma (T^4 - T_0^4) \quad (11)$$

$$q_{evp} = m_v L_v \quad (12)$$

where Q_{abs} is the absorbed laser energy, $\vec{n}$ is the normal vector of the free interface of the molten pool, q_{cov} , q_{rad} , and q_{evp} are respectively the heat convection energy loss, the heat radiation energy loss, and the evaporation energy loss, h_c is the convection heat transfer coefficient, ϵ is the heat radiation coefficient, σ is the Stephen-Boltzmann constant, m_v and L_v are the evaporation rate and evaporation latent of the liquid metal, respectively.

3.3.2. Momentum boundary condition

$$P_L = F_S + P_V + P_M + F_V \quad (13)$$

$$F_S = \gamma_m + \frac{d\gamma}{dT} (T - T_m) \quad (14)$$

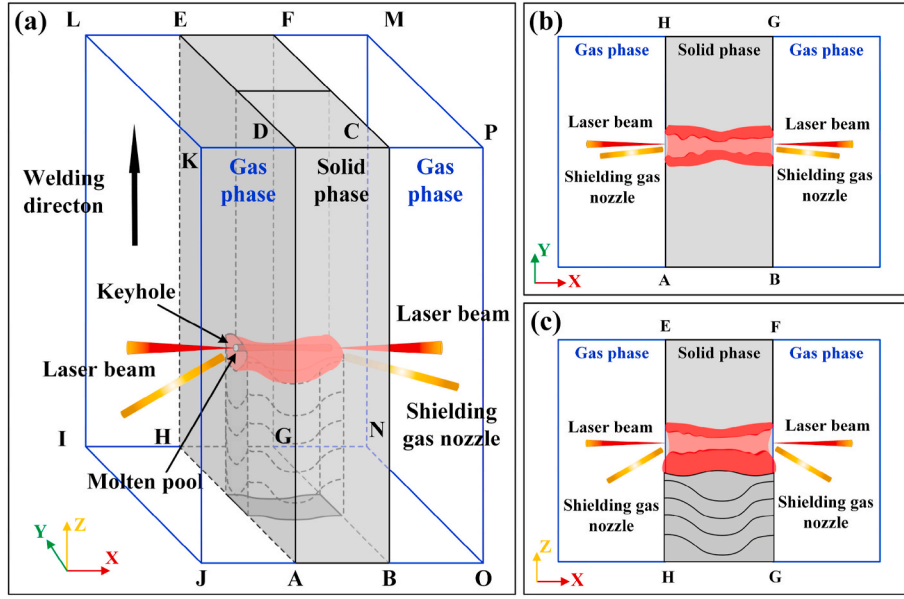


Fig. 2. Solution model and boundary conditions for calculation in vertical-up laser mirror welding: (a) Schematic diagram of boundary conditions as a whole, (b) Local boundary conditions on the X-Y section, and (c) Local boundary conditions on the X-Z section.

$$P_V = \alpha B_0 T_W^{-\frac{1}{2}} \exp\left(\frac{-M_a L_V}{N_a \sigma T_W}\right) \quad (15)$$

where P_L is the total pressure, F_s is the interfacial surface tension, P_V is the vapor recoil pressure, P_M is the hydrostatic pressure of molten pool, F_V is the hydrodynamic pressure of molten pool, α is the gas pressure coefficient, B_0 is the evaporation constant of material [35], T_W is the temperature of keyhole wall, M_a is the molar mass, N_a is the Avogadro constant, γ_m is the surface tension at melting point temperature, and T_m is the melting point of material. Among them, the vapor recoil pressure and surface tension are the main driving forces in the flow process of molten pool, which have an important impact on the fluid flow of molten pool.

3.4. Heat source model

The surfaces on both sides of BM are heated and melted by two symmetrical laser heat sources simultaneously during the vertical-up laser mirror welding process. The schematic diagram of heat source model used is presented in Fig. 3. Part of the laser input could be equivalent to a surface heat source due to the formation of high-energy plasma on the molten pool surface [36]. While the laser energy is absorbed by the inverse bremsstrahlung and Fresnel absorption mechanism in the molten pool, accompanied by the continuous evaporation of liquid metal in the keyhole wall. It is indicated that the keyhole full of metallic steam could be regarded as a body heat source with a certain depth [37,38]. Therefore, a combined heat source model is used to simulate the energy transfer of laser beam in this paper. The energy

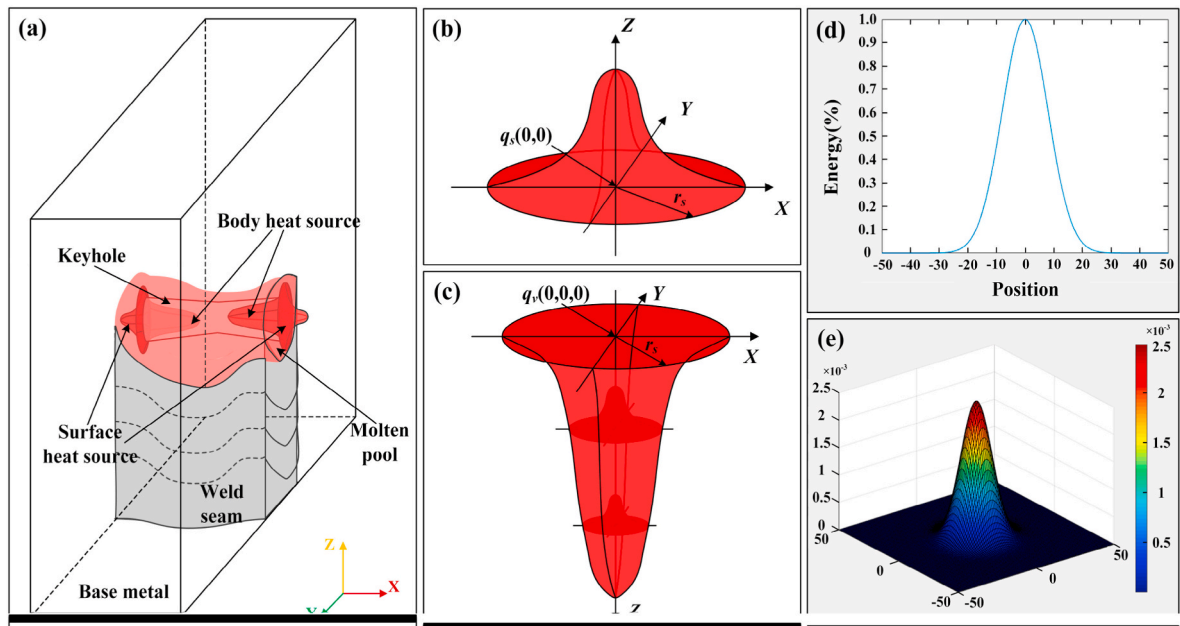


Fig. 3. Schematic diagram of heat source model: (a) Schematic diagram of heat source zone, (b) Surface heat source, (c) Body heat source, (d) Two-dimensional energy distribution, and (e) Three-dimensional energy distribution.

distribution of two-dimensional and three-dimensional laser Gaussian heat source is shown in Fig. 3(d) and (e).

The surface heat source model is depicted as follows:

$$q_s(x, y) = \frac{\eta_s Q_s}{\pi r_s^2} \exp\left(-\frac{\alpha_r(x^2 + y^2)}{r_s^2}\right) \quad (16)$$

where η_s is the concentration coefficient of the surface heat source, Q_s is the effective surface heat source energy, α_r is the correction coefficient and r_s is the heat source effective radius.

The body heat source model is written as follows:

$$q_v(x, y, z) = \frac{9\eta_v Q_v}{\pi H r_s^2 (1 - e^{-3})} \exp\left(\frac{-9(x^2 + y^2)}{r_s^2 \ln(H/z)}\right) \quad (17)$$

where η_v is the concentration coefficient of the body heat source, Q_v is the effective energy of body heat source. H is the effective depth of body heat source.

In addition, the surface heat source and body heat source energy both follow the relationship with the total laser energy:

$$\eta Q = Q_s + Q_v \quad (18)$$

where η is the effective absorption coefficient. Q is the energy of laser acting on the base metal [39]. Among them, r_s is 0.325 mm and H is 3 mm.

3.5. Coordinate system transformation

According to the spatial characteristics of the symmetrical heat sources applied on the left and right sides during the vertical-up laser mirror welding process, the coordinates of heat sources on both sides were transformed accordingly. The coordinates of the left and right heat sources after conversion can be expressed as follows:

$$\begin{cases} x_l = y' + a \\ y_l = -(x' + b) \\ z_l = z' + c \end{cases} \quad (19)$$

$$\begin{cases} x_r = -(y' + a) \\ y_r = x' + b \\ z_r = z' + c \end{cases} \quad (20)$$

where $x_l, y_l, z_l, x_r, y_r, z_r$ and x', y', z' are the spatial coordinates of the left heat source and the right heat source after conversion respectively, x', y', z' are the coordinate origin of the model, a, b and c are the offsets in the x, y and z directions respectively in the local coordinate system.

3.6. Keyhole surface tracking

A multi-phase VOF method was used to track the flow physics of liquid-gas interface [40]. The equation of the method is drawn as follows:

$$\frac{\partial T}{\partial t} + \nabla \cdot (v_L F) = 0 \quad (21)$$

where v_L is fluid velocity, F is a function of x, y, z , and t , which is the volume fraction of the cell. If $F = 0$, the domain is occupied by gas phase; If $F = 1$, the domain is occupied by metal liquid; If $0 < F < 1$, the domain is located on the phase interface which includes both liquid and gas [41, 42]. The phase interface of liquid-gas is shown in Fig. 4.

3.7. Computational domain and material properties

The calculation model with an overall size of 20 mm × 20 mm × 18 mm is simplified into three identical regions, in which the model size corresponding to metal phase and gas phase on both sides is 20 mm × 20 mm × 6 mm. The details of dimensions and mesh are shown in Fig. 5.

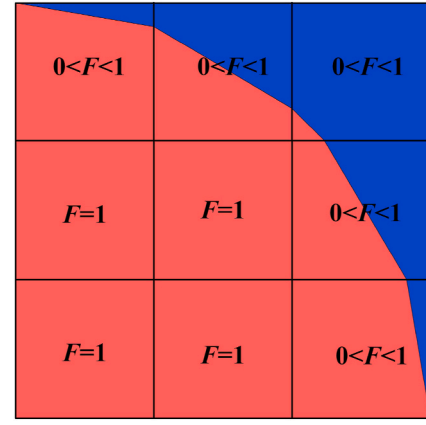


Fig. 4. Schematic diagram of the liquid-gas interface.

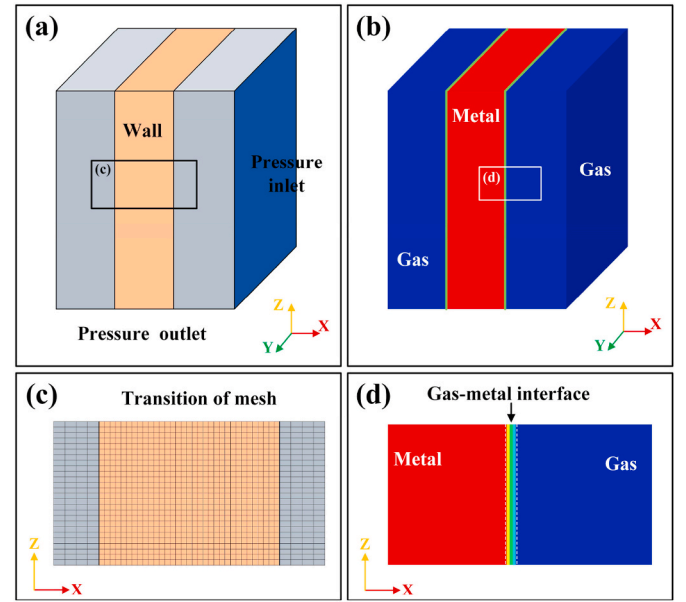


Fig. 5. Mesh model and calculation domain of the vertical-up laser mirror welding: (a) Model of the calculation domain, (b) Calculation domain with phase distribution, (c) Partial of mesh model, and (d) Schematic diagram of gas-metal interface marked in (b).

The mesh of weld seam (WS) is optimized, while the mesh size between different regions is consistent to ensure smooth transition between meshes in each region. The mesh size of optimized area is 0.15 mm in x, y and z directions, while the mesh size of other area is 0.3 mm. In this paper, the ultimate mesh model is divided into 637,760 hexahedral mesh elements. The main thermal physical properties of BM utilized in this paper are given in Table 3 [43].

4. Results and discussion

4.1. Comparison with the simulation and experimental data

Computational fluid dynamics (CFD) simulation modeling and VOF multiphase model have been adopted in the present study. Fig. 6 shows the comparison results between simulation and experiment under the process parameters given in Table 2, where the temperature contour represents the simulated weld profile. Fig. 6 shows the simulation results of the fusion zones in the transverse and longitudinal cross sections after the solidification of the molten pool. Although the weld morphology of calculated results is slightly different from the experimental results, the

Table 3

Thermophysical parameters of 2219 aluminum alloy.

Name	Symbol	Unit	Value
Solidus temperature	T_s	K	820
Liquid temperature	T_l	K	930
Thermal conductivity of solid	k_s	$\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$	238
Thermal conductivity of liquid	k_l	$\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$	100
Solidus density	ρ_l	$\text{kg}\cdot\text{m}^{-3}$	2700
Liquid density	ρ_s	$\text{kg}\cdot\text{m}^{-3}$	2400
Boiling point	T_b	K	2730
Specific heat of solid	C_{ps}	$\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$	871
Specific heat of Liquid	C_{pl}	$\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$	1060
Latent heat of fusion	H_m	$\text{J}\cdot\text{kg}^{-1}$	3.87×10^{-5}
Latent heat of evaporation	H_v	$\text{J}\cdot\text{kg}^{-1}$	1.08×10^{-7}
Dynamic viscosity	μ	$\text{kg}\cdot\text{m}^{-1}\cdot\text{s}^{-1}$	0.006
Coefficient of thermal expansion	β	K^{-1}	2.36×10^{-5}
Convective heat transfer coefficient	h_c	$\text{W}\cdot\text{K}^{-1}\cdot\text{m}^{-2}$	15
Thermal emissivity	ε		0.4
Ambient temperature	T_{ref}	K	300
Surface tension	σ	$\text{N}\cdot\text{m}^{-1}$	0.914
Temperature gradient of surface tension	A_σ	$\text{N}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$	-0.35×10^{-3}

figures indicate that the numerical model accurately represents energy transfer during the vertical-up laser mirror welding process. For the calculated fusion zone in Fig. 6, the penetration depth and weld width are predicted considerably accurately. Compared with the experimental results, the fusion zone is a little wider at the middle region, where the width of the simulation results is 2.70 mm, while the width of the experimental results is 2.57 mm, owing to the neglect of the energy which loses from the penetrated keyhole [20,44,45]. Besides, the results shows that the simulated WS is well matched with the experimental WS in sizes, which verifies the validity of the developed model.

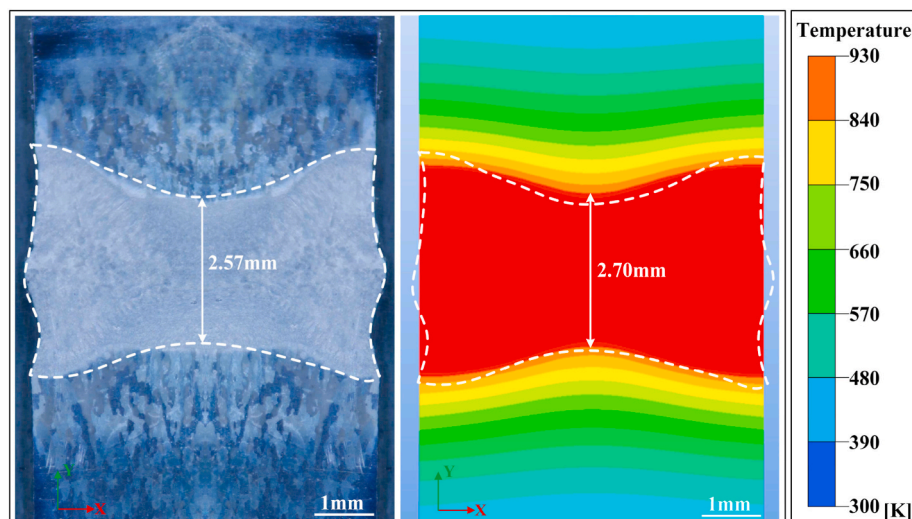
4.2. Flow behavior of molten pool

Fig. 7 shows the morphology and flow field of molten pool on the X-Y and X-Z sections before keyhole coupling. Even if the same process parameters are utilized on both sides during the vertical-up laser mirror welding process, it is difficult to ensure that the upper and lower molten pool as well as the left and right keyholes are completely consistent in shape within a certain period of time. As the welding process proceeds, the fluid velocity of molten pool decreases from 5.26 m s^{-1} to 2.151 m s^{-1} . The fluid with the maximum velocity of 5.26 m s^{-1} appears on the surface of molten pool in the middle of keyholes when $t = 15 \text{ ms}$, which indicates that the diameter of keyholes tends to increase. At this

moment, Marangoni flow occurs on the molten pool zone around the surface of keyhole [46]. While on the X-Z section, the morphology of upper and lower molten pool has a significant difference because of the effect of gravity and the confused high velocity fluid on the keyhole wall is formed when $t = 15 \text{ ms}$. When the welding process reaches $t = 25 \text{ ms}$, it can be clearly seen that the flow direction of upper and lower molten pool becomes consistent. Besides, the morphology and flow field of upper and lower molten pool both exhibits better symmetry and consistency on the X-Y section.

Fig. 8 shows the morphology and flow field of molten pool on the X-Y and X-Z sections during and after keyhole coupling. Compared with that before keyhole coupling, the flow velocity of molten pool during keyhole coupling is obviously decreased, and the maximum velocity in only 1.384 m s^{-1} , as shown in Fig. 8(a)–(b). At this time, Marangoni flow is found in the molten pool zone at the front end of coupled keyhole, resulting in an inconsistent flow field of upper and lower molten pool on the X-Y section. Besides, the keyhole diameter in the center zone is very small when keyhole on both sides is just coupled. As the welding process proceeds, the keyhole diameter at each position indicates an increasing trend. At $t = 38 \text{ ms}$, a high velocity fluid is developed on the surface of upper and lower molten pool in the keyhole coupling center zone of X-Y section in Fig. 8(c), which can increase the keyhole diameter at this zone and enable keyhole coupling stability. While on the X-Z section in Fig. 8(d), the high velocity is achieved on the surface of lower molten pool. Consequently, a downward motion trend occurs for the keyhole coupling zone. After keyhole coupling tends to be stable at $t = 60 \text{ ms}$ in Fig. 8(e), the flow field of upper and lower molten pool also exhibits better symmetry and consistency on the X-Y section. On the X-Z section at $t = 60 \text{ ms}$ in Fig. 8(f), the flow field of upper and lower molten pool in the keyhole coupling center zone is affected by gravity and flows downward, while the surface of molten pool has an upward flow tendency under surface tension and other forces [47]. With the movement of the coupled keyhole, the morphology of upper and lower molten pool has almost no change, and the keyhole diameter in the center zone gradually increases, which improves the stability of the coupled keyhole. In addition, after keyhole coupling process is completed at $t = 150 \text{ ms}$ in Fig. 8(g)–(h), a firm connection is finally formed.

Fig. 9 shows the variation in morphology extraction and area of molten pool with time on the X-Y and X-Z sections. Fig. 9(a) shows the schematic diagram of observation position. Fig. 9(b) and (c) are the molten pool extraction in Plane b and Plane c in Fig. 9(a), respectively. It can be clearly seen from Fig. 9(b) and (c) that a small discrepancy is raised between the morphology of upper and lower molten pool on the X-Y section, while a significant difference is appeared between the

**Fig. 6.** Validation of the numerical model.

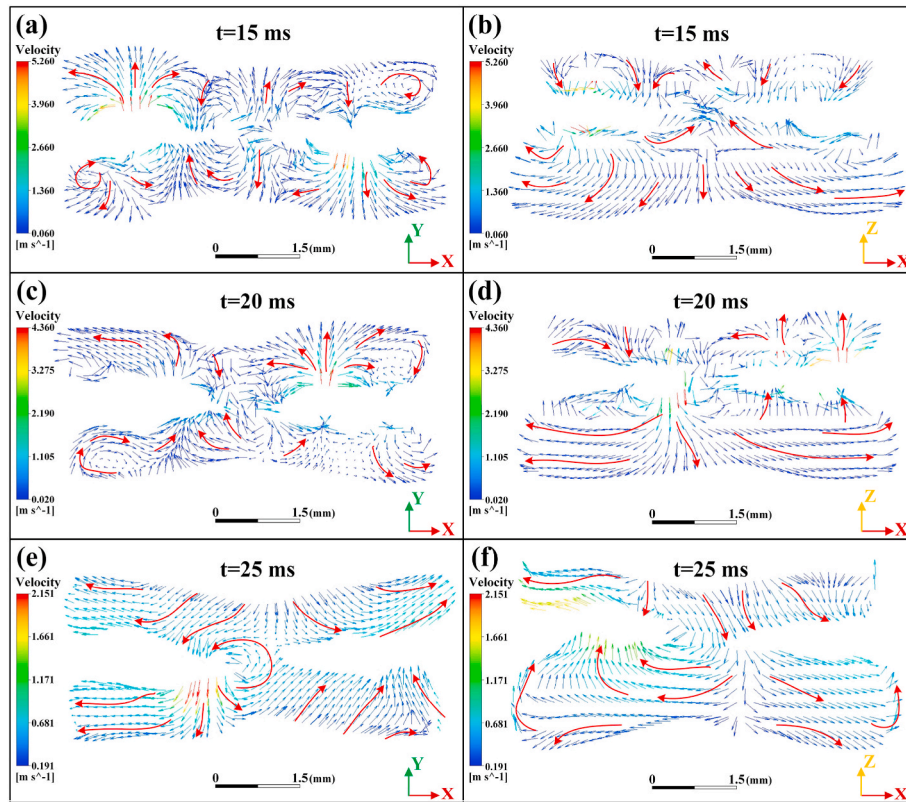


Fig. 7. The morphologies and flow field of molten pool on X-Y and X-Z sections before keyhole coupling: (a)–(b) $t = 15$ ms, (c)–(d) $t = 20$ ms, and (e)–(f) $t = 25$ ms.

morphology of upper and lower molten pool on the X-Z section. Hence, the flow behavior of molten pool in the vertical-up laser mirror welding process is quantitatively analyzed by extracting the area of upper and lower molten pool. On the X-Y section in Fig. 9(d), the total area of molten pool gradually increases with time, and the same trend is also observed for the area of upper and lower molten pool. Moreover, the small discrepancy of area between the upper and lower molten pool accompanies the process before keyhole coupling. Prominently, during the keyhole coupling occurs from 30 ms to 40 ms, the difference between the upper and lower molten pool can be almost ignored. After that, the keyhole tends to be stable and the difference of area between the upper and lower molten pool is almost unchanged. While on the X-Z section in Fig. 9(e), the area change of upper and lower molten pool is apparently different from that of the X-Y section. The area of lower molten pool decreases due to the effect of Marangoni flow during the keyhole coupling, while the area of upper molten pool decreases between 60 ms and 100 ms due to the effect of gravity and then remains unchanged with the stability of the coupled keyhole.

4.3. Dynamic evolution characteristics of keyhole

Fig. 10 shows the morphology of keyhole wall before coupling. As a result of the vapor recoil pressure, the liquid metal in molten pool heated by laser beam is evaporated rapidly and keyholes are formed. The temperature of keyhole wall is significantly higher than that of the surrounding zone. It can be clearly seen in Fig. 10(a)–(d) that the rough keyhole wall on both sides is found before coupling, which indicates that a certain degree of keyhole wall fluctuation is captured on the left and right keyholes, thus a slight difference of the morphology between the left and right keyhole is caused and humps and wrinkle are formed on the keyhole wall due to the fluctuation. The fluctuation is caused by the Fresnel absorption effect, which makes the surface temperature of keyhole rise sharply and the metal evaporation become violent.

The flow field of keyhole wall before coupling is further analyzed in

Fig. 11. Compared to the results in Fig. 10(a)–(d), the calculated distribution characteristics of flow field at each moment are relatively similar. At $t = 15$ ms, the fluid velocity distribution on both sides of the keyhole wall differs greatly in Fig. 11(a). The maximum velocity zone is distributed in the middle of left keyhole wall, and the maximum velocity reaches 5.260 m s^{-1} , which is the same as the flow field of molten pool in Fig. 7(a). Therefore, the fluid flow behavior on the keyhole wall is closely related to the fluid flow in the molten pool. Besides, the maximum velocity zone is also located at the position where the hump is formed, which indicates that the hump is closely relevant to the fluid velocity at this time. It can be proved that the increasing trend of keyhole diameter is improved by the high velocity fluid. As the welding process continues, the fluid velocity of molten pool decreases from 5.260 m s^{-1} to 1.925 m s^{-1} . The fluid on the keyhole wall is characterized by flowing from the bottom of keyhole along the wall to the opening area, as shown in Fig. 11(c)–(d).

Fig. 12 shows the morphology of keyhole during and after coupling. In the process of keyhole coupling, the left and right keyhole both are tapered and the diameter of keyhole in the coupling central zone is obviously different from that of other places. Specially, the diameter of keyhole in the coupling central zone is the smallest when keyholes are just coupled. This indicates a weak zone and is vulnerable to the influence of other factors, resulting in occurrence of humps in this stage. As presented in Fig. 12(a)–(d), it is acquired that the temperature of keyhole in the coupling central zone is slightly lower than that in other zones of keyhole. The uneven temperature distribution might be caused by the energy attenuation of the laser beam and the fluctuation of keyhole at the initial stage of coupling. Moreover, the keyhole in the coupling central zone is dragged by the gravity between $t = 32$ ms and $t = 38$ ms and a downward hump is formed. It can be concluded that the fluctuation still exists in the coupling keyhole at different times during the vertical-up laser mirror welding process and the fluctuation degree of keyhole in the coupling central zone is more significant than that in other zones. This might be because the bottom of the keyhole is subject

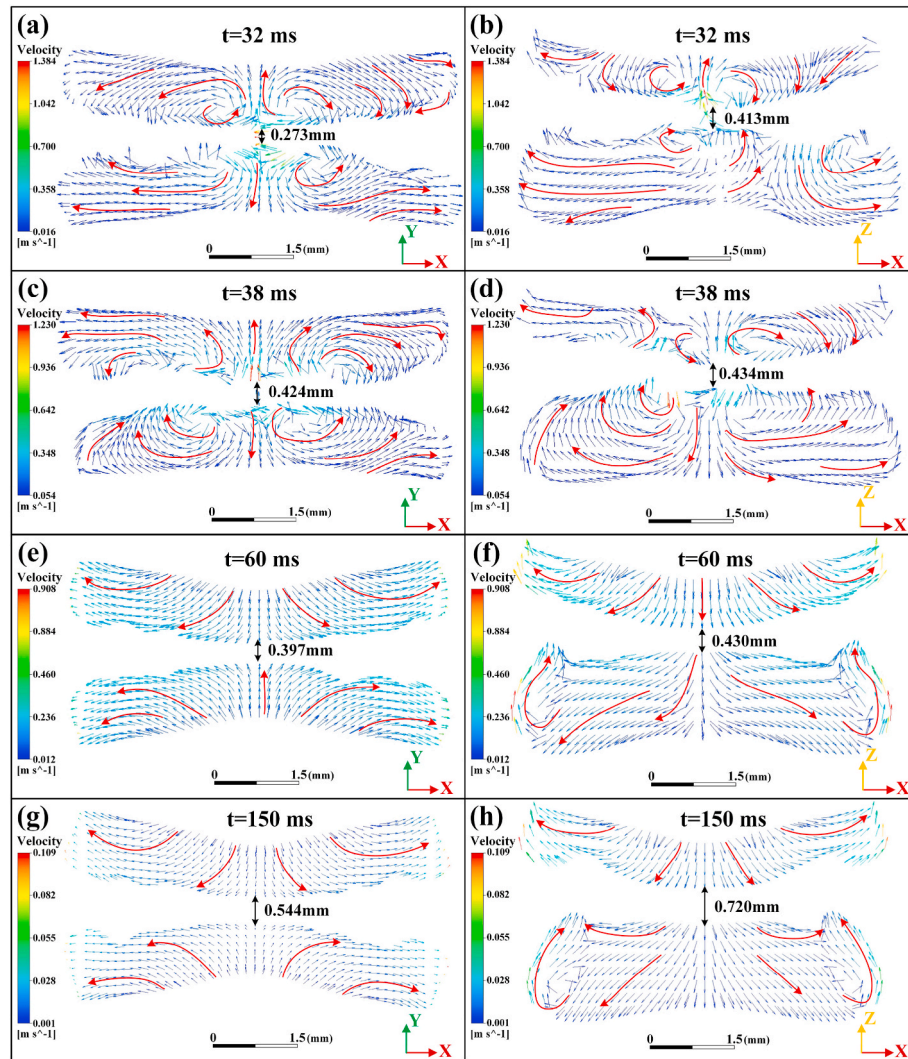


Fig. 8. The flow field of molten pool on XY and XZ sections during and after keyhole coupling: (a)–(b) $t = 32$ ms, (c)–(d) $t = 38$ ms, (e)–(f) $t = 60$ ms, and (g)–(h) $t = 100$ ms.

to the common effect of single beam laser energy attenuation and double beam laser energy accumulation, which makes the energy change in this zone more violent, thus a large fluctuation of the keyhole morphology is caused. As the welding process progresses, the diameter of keyhole in the coupling central zone gradually increases in Fig. 12(e)–(h) and the coupled keyhole tends to become stable. Finally, a firm connection is according formed.

For further analyzing the process of keyhole coupling and stabilization, the flow field during and after keyhole coupling is observed in Fig. 13. In the process of keyhole coupling and stabilization, the maximum flow velocity at each moment varies from 1.384 m s^{-1} to 0.109 m s^{-1} , which indicates the stability of the coupled keyhole gradually increases. It is presented in Fig. 13(a)–(c) that the surrounding fluid is distributed in the opening zone of keyhole on both sides and the high velocity fluid is appeared in the coupling central zone, as well as a fluid flow characteristic along the keyhole wall is found at the bottom of the coupled keyhole. At $t = 40$ ms, a confused high velocity fluid on the keyhole wall is formed and the fluid flow direction in the coupling keyhole becomes consistent until $t = 60$ ms. As the welding process continues to the coupled keyhole becoming stable, the maximum velocity is only 0.109 m s^{-1} at this time in Fig. 13(f). The fluid velocity of keyhole is lowest on the keyhole wall, which indicates that the coupled keyhole has been stable while the fluid velocity in the keyhole opening zone is large due to the movement of laser heat source.

In order to reveal the keyhole evolution during the vertical-up laser mirror welding, the area of molten pool and keyhole was extracted, as shown in Fig. 14. Fig. 14(a) and (b) respectively show the exaction of uncoupled keyhole and coupled keyhole. The rough keyhole wall is formed before coupling and the keyhole becomes smooth after coupling, which indicates that the keyhole gradually becomes stable. As shown in Fig. 14(c), the total area of left and right keyhole gradually increases with time before keyhole coupling. During the time period when keyhole coupling occurs between $t = 30$ ms and $t = 40$ ms, the total area of keyhole changes little, but the area of molten pool increases rapidly from 11.1 mm^2 to 12.9 mm^2 . After keyhole coupling, that is, between 40 ms and 60 ms, there is slight increase of keyhole and molten pool area since the keyhole is just coupled in this period and a fluctuating through keyhole is formed inside. When the keyhole coupling process is stabilized (after about 100 ms), the area of molten pool increases significantly while that of coupled keyhole changes little, indicating that keyhole coupling has a significant improvement in the melting efficiency of laser. The area difference between the two keyholes before coupling are shown in Table 4, in which the area difference between the left and right keyhole does not exceed 0.06 mm^2 , implying that comparable symmetry could be visible during the vertical-up laser mirror welding process.

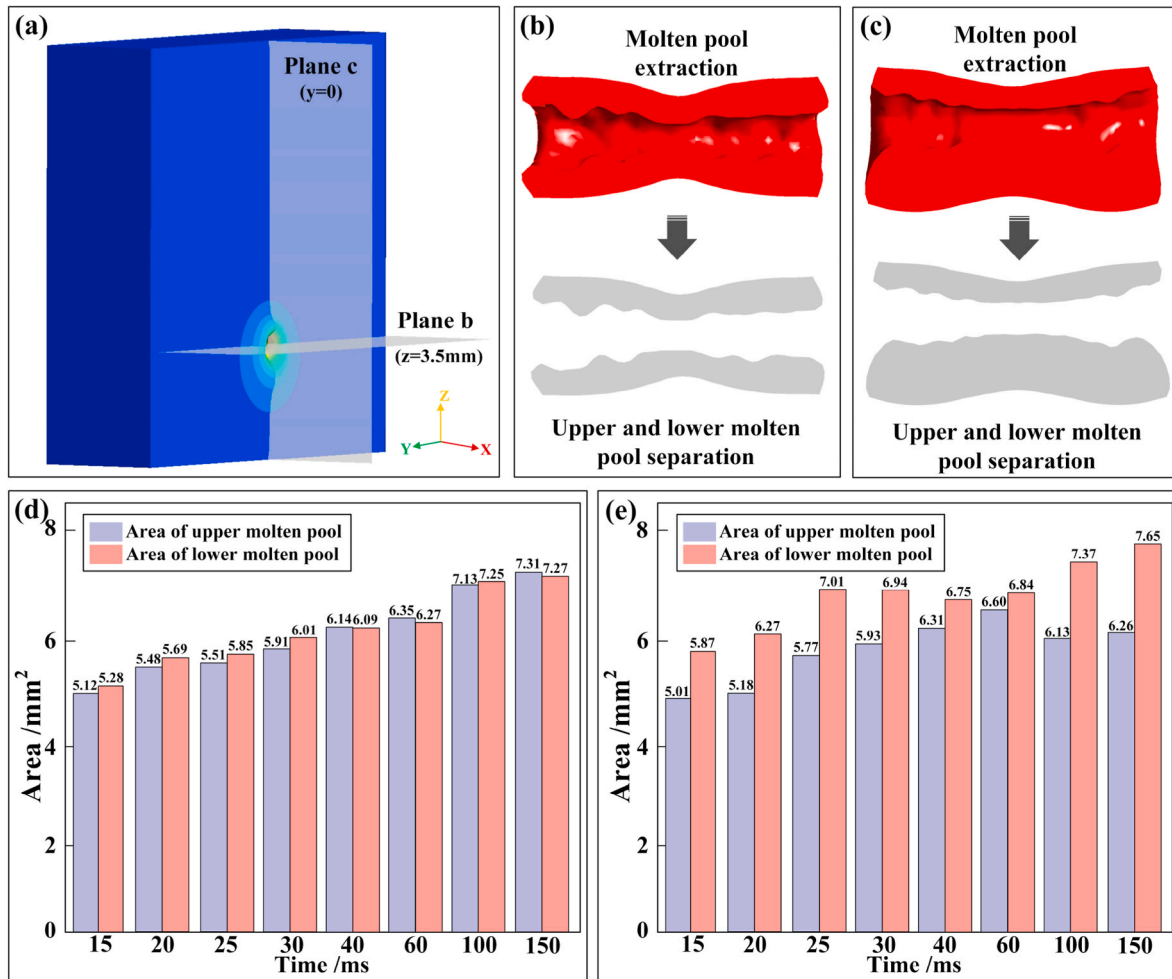


Fig. 9. Morphology extraction and area of upper and lower molten pool on X-Y and X-Z sections: (a) Schematic diagram of observation position, (b) Molten pool extraction in Plane b, (c) Molten pool extraction in Plane c, (d) Area of upper and lower molten pool in (b), and (e) Area of upper and lower molten pool in (c).

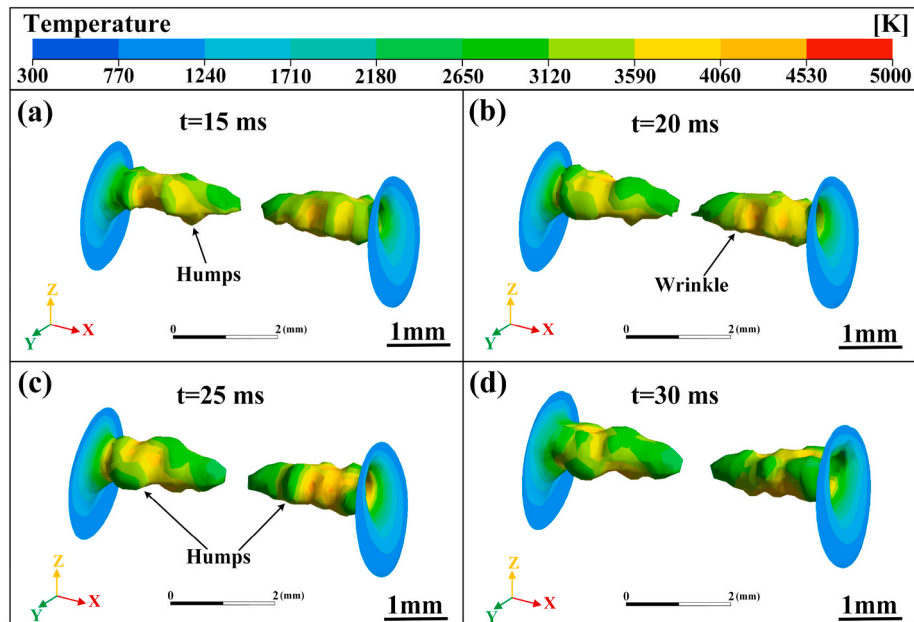


Fig. 10. The morphology of keyhole before coupling: (a) $t = 15$ ms, (b) $t = 20$ ms, (c) $t = 25$ ms, and (d) $t = 30$ ms.

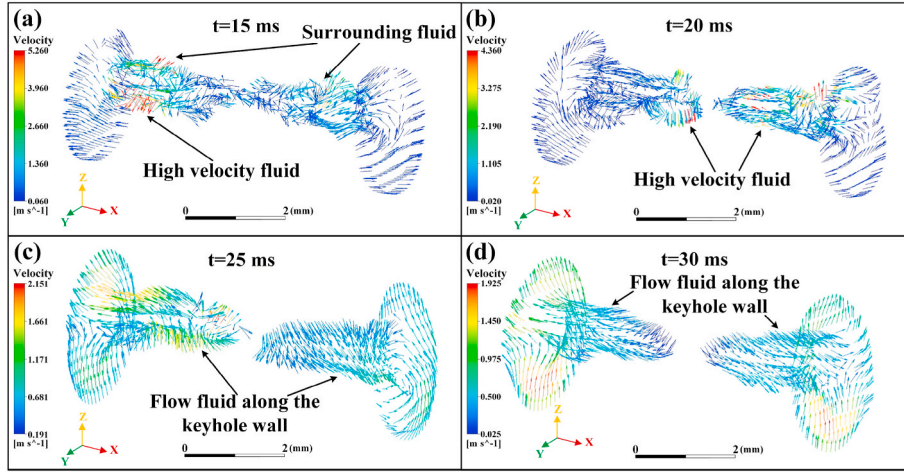


Fig. 11. The flow field of the keyhole before coupling: (a) $t = 15$ ms, (b) $t = 20$ ms, (c) $t = 25$ ms, and (d) $t = 30$ ms.

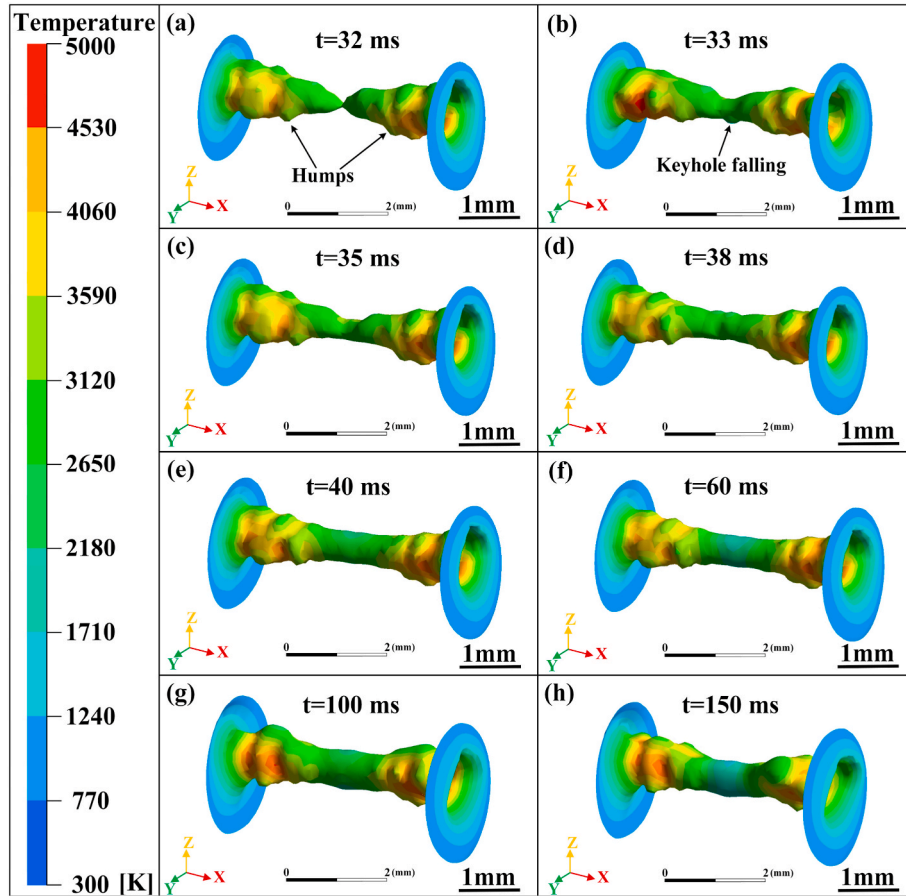


Fig. 12. The morphology of keyhole during and after coupling: (a) $t = 32$ ms, (b) $t = 33$ ms, (c) $t = 35$ ms, (d) $t = 38$ ms, (e) $t = 40$ ms, (f) $t = 60$ ms, (g) $t = 100$ ms, and (h) $t = 150$ ms.

4.4. Force analysis of keyhole

Fig. 15 shows the force situation before keyhole coupling on the X - Y and X - Z sections. During the evolution of keyhole, the local position of keyhole wall is affected by friction force (F_f), surface tension (F_s), recoil pressure (P_v), hydrodynamic pressure (F_v), hydrostatic pressure (P_m) and gravity (g). Among them, only the X - Z section is affected by gravity. At the bottom of keyhole, the force conditions at points A, B, C and D are:

Point A:

$$F_{vx} + P_{mx} < F_{fx} + F_{sx} + P_{vx} \quad (22)$$

$$F_{fy} + F_{sy} + P_{my} < P_{vy} + F_{vy} \quad (23)$$

Point B:

$$F_{vx} + P_v > P_m \quad (24)$$

$$F_f + F_s < F_{vy} \quad (25)$$

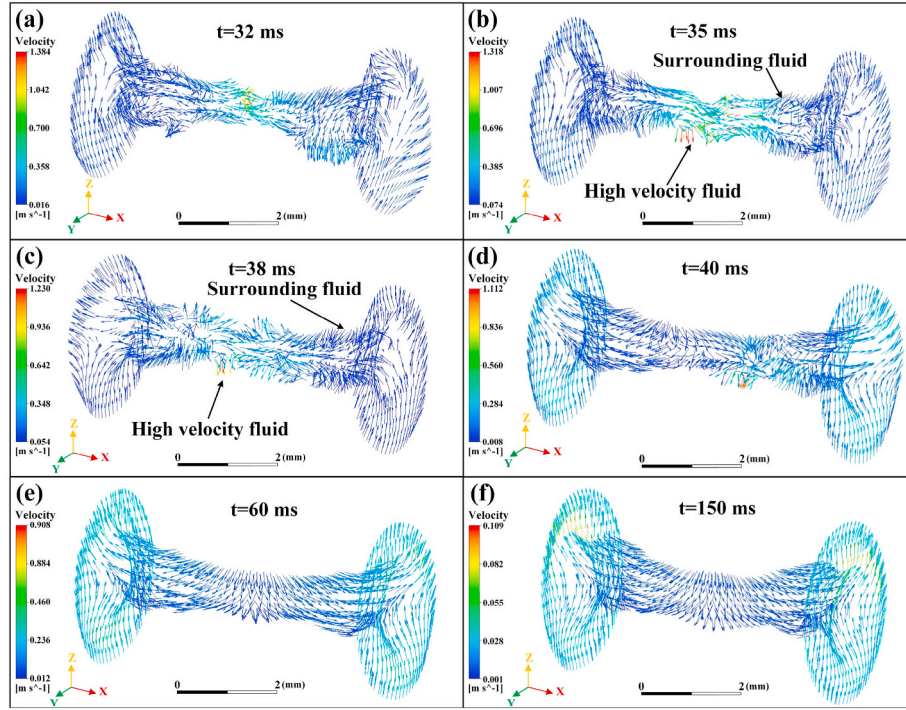


Fig. 13. The flow field of the keyhole during and after coupling: (a) $t = 32$ ms; (b) $t = 35$ ms; (c) $t = 38$ ms, (d) $t = 40$ ms, (e) $t = 60$ ms, and (f) $t = 150$ ms.

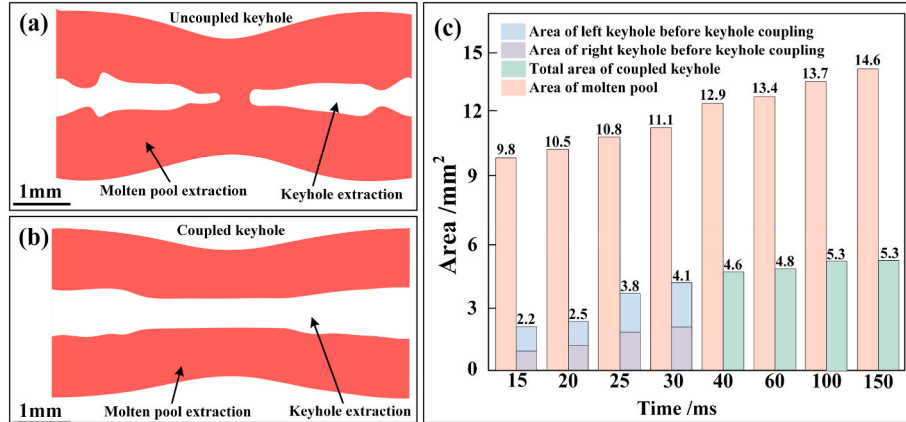


Fig. 14. Extraction and area of keyhole in the vertical-up laser mirror welding process: (a) Extraction of uncoupled keyhole, (b) Extraction of coupled keyhole, and (c) Area of the keyhole before and during keyhole coupling.

Table 4

Cross sectional area of the left and right keyhole before keyhole coupling.

Time (ms)	Area of left keyhole (mm ²)	Area of right keyhole (mm ²)	Total area of keyhole (mm ²)	Difference between left and right keyhole area (mm ²)
15	1.14	1.16	2.30	0.02
20	1.19	1.15	2.34	0.04
25	1.64	1.58	3.22	0.06
30	1.82	1.78	3.60	0.04

Point C:

$$F_{vx} + P_{mx} < F_{fx} + F_{sx} + P_{vx} \quad (26)$$

$$F_{fc} + F_{sz} + P_{mz} < P_{vz} + F_{vz} + g \quad (27)$$

Point D:

$$F_{vx} + P_{vx} > P_{mx} + F_{sx} + F_{fx} \quad (28)$$

$$F_{fc} + F_{sz} + P_{vz} < F_{vz} + P_{mz} + g \quad (29)$$

Where F_{vx} , P_{mx} , F_{fx} , F_{sx} , P_{vx} , F_{vy} , P_{my} , F_{fy} , F_{sy} , P_{vy} , F_{vz} , P_{mz} , F_{fz} , F_{sz} and F_{sz} are the X-axis, Y-axis and Z-axis components of F_v , P_m , F_f , F_s and P_v respectively. According to the force condition at points A, B, C and D, the direction of the resultant force on the X-axis is consistent with the depth direction of keyhole. Therefore, the depth of keyhole will continue to increase until a through keyhole is formed, as shown in Fig. 11. On the X–Y section, the resultant force direction of point A and point B on the Y-axis is consistent with the radial direction of keyhole, and both have the trend of increasing the diameter of keyhole, but the direction is opposite. On the X–Z section, the resultant force of points C and D on the Z-axis points downward, so the keyhole will eventually be coupled at the lower position.

Fig. 16 describes the force analysis of local position during keyhole

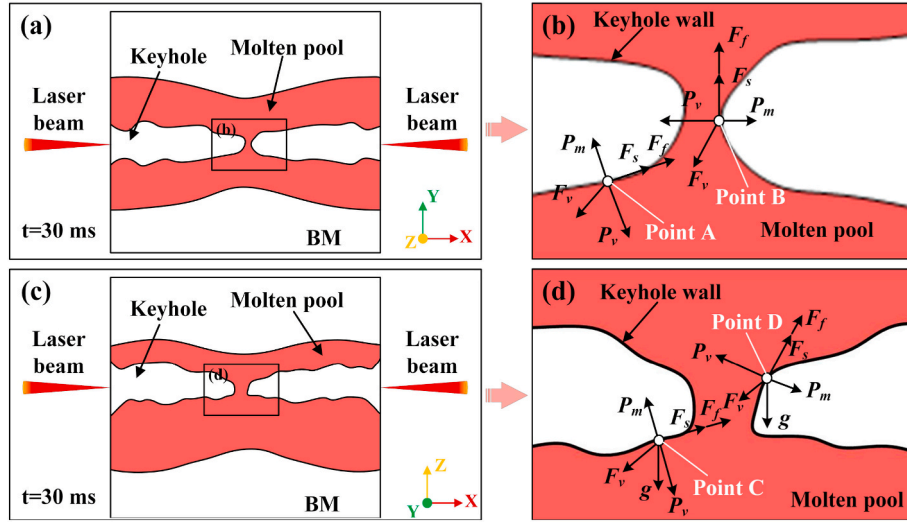


Fig. 15. The force analysis of keyhole before keyhole coupling: (a) Schematic diagram of keyhole on X-Y section before coupling, (b) The forces analysis on the tip of keyhole, (c) Schematic diagram of keyhole on X-Z section before coupling, and (d) The forces analysis on the tip of keyhole.

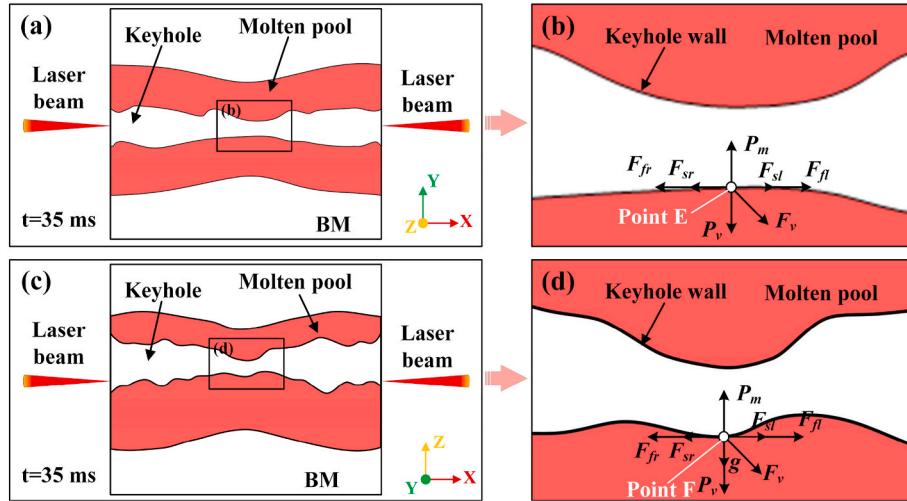


Fig. 16. The force analysis of keyhole during keyhole coupling: (a) Schematic diagram of keyhole on X-Y section during coupling, (b) The forces analysis of keyhole wall, (c) Schematic diagram of keyhole on X-Z section during coupling, and (d) The forces analysis of keyhole wall.

coupling on the X-Y and X-Z sections. It can be seen that the force condition of keyhole has changed significantly during keyhole coupling. On the lower wall of keyhole coupling zone, the force condition at points E and F is:

Point E:

$$F_{fr} + F_{sr} < F_{sl} + F_{fl} + F_{vx} \quad (30)$$

$$P_y + F_{vy} > P_m \quad (31)$$

Point F:

$$F_{fr} + F_{sr} < F_{sl} + F_{fl} + F_{vx} \quad (32)$$

$$P_y + F_{vz} + g > P_m \quad (33)$$

Where F_{fl} , F_{fr} , F_{sl} and F_{sr} are F_f and F_s of the left keyhole and the right keyhole respectively. According to the force condition on the X-axis at the central zone of keyhole, the keyhole on both sides will be continuously coupled. On the X-Y and X-Z sections, the resultant force of points E and F points downward, but the resultant force of point E is the direction of increasing the diameter of keyhole, which is conducive to

improving the stability of keyhole, while the resultant force of point F makes the keyhole coupling center zone falling as shown in Fig. 12(c).

Fig. 17 shows the force analysis of keyhole after keyhole coupling. On the upper wall of keyhole, the force condition at points G and H is:

Point G:

$$F_{fr} + F_{sr} + F_{vx} > F_{sl} + F_{fl} \quad (34)$$

$$P_y + F_{vy} > P_m \quad (35)$$

Point H:

$$F_{fr} + F_{sr} + F_{vx} > F_{sl} + F_{fl} \quad (36)$$

$$P_y + F_{vz} > P_m + g \quad (37)$$

According to the force condition on the X-axis at the central zone of keyhole, the keyholes on both sides will be continuously coupled. For the upper wall of keyhole on the X-Y and X-Z sections, the resultant force of points G and H is consistent with the radial direction of keyhole. So, both are beneficial to increase the diameter of keyhole and improve the stability of keyhole.

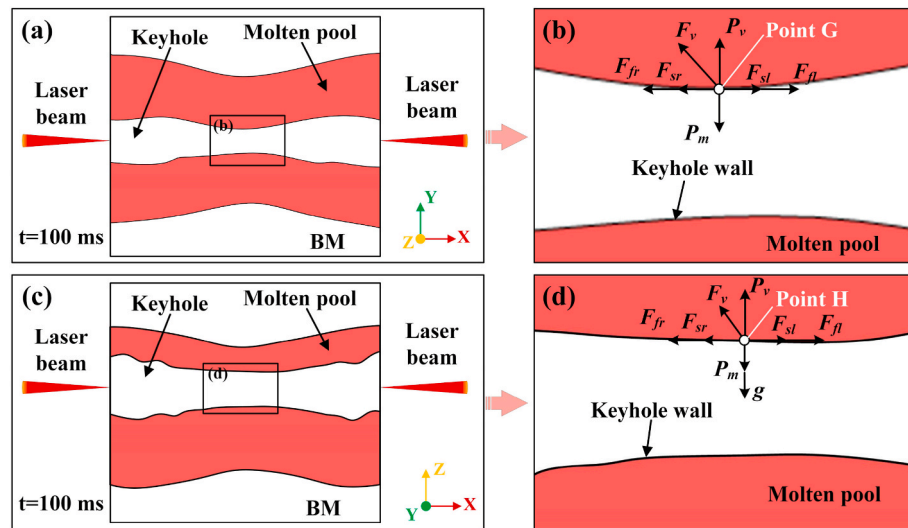


Fig. 17. The force analysis of keyhole after keyhole coupling: (a) Schematic diagram of keyhole on X–Y section after coupling, (b) The forces analysis of keyhole wall, (c) Schematic diagram of keyhole on X–Z section after coupling, and (d) The forces analysis of keyhole wall.

5. Conclusion

The molten pool flow behavior and keyhole dynamic characteristics of 2219 aluminum alloy on the X–Y and X–Z sections during the vertical-up laser mirror welding were investigated by CFD simulation in the present study. Based on the model verification, the evolution behavior and force analysis of molten pool and keyhole were discussed. The main conclusions were summarized as follows.

- (1) The calculated values indicates that the numerical model accurately represents heat transfer during the vertical-up laser mirror welding process and the simulation results are basically consistent with experiment data, which verifies the validity of the developed model.
- (2) As the keyhole on both sides varies from before coupling to when coupling, the maximum fluid velocity of molten pool gradually decreases from 5.260 m s^{-1} to 0.109 m s^{-1} and its direction is changed from confusion to consistent. Compared with that on the X–Z section, high degree of consistency for the morphology and flow field distribution of upper and lower molten pool is exhibited better consistency on the X–Y section.
- (3) When the keyhole on both sides is just coupled, the diameter of keyhole in the coupling central zone is the smallest, which is vulnerable to the influence of other factors and results in occurrence of humps at this stage. Besides, the fluctuation degree of keyhole in the coupling central zone is more significant than that in other zones.
- (4) On the X–Y section, the total area of molten pool gradually increases with time and the same trend is also observed for the area of upper and lower molten pool. The total area of left and right keyhole gradually increases with time before keyhole coupling, and during the time period when keyhole coupling occurs between $t = 30 \text{ ms}$ and $t = 40 \text{ ms}$, the total area of keyhole changes little while the area of molten pool increases rapidly from 11.1 mm^2 to 12.9 mm^2 .
- (5) As the welding process progresses, the diameter of keyhole in the coupling central zone significantly increases under the resultant force and the coupled keyhole tends to become stable, finally forming a firm connection.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

This work is supported by the Natural Science Foundation of China (Grant No. 52205380), the Natural Science Foundation of Jiangsu Province of China (Grant No. BK20220900), the Natural Science Foundation of Liaoning Province of China (Grant No. 2021-BS-050), the China Postdoctoral Science Foundation (Grant No. 2022M721595), the Fundamental Research Funds for the Central Universities (Grant No. NS2022062), the State Key Laboratory of Advanced Welding and Joining, Harbin Institute of Technology (Grant No. AWJ-23M04), and the Scientific Research Foundation for Introduced Talents of Nanjing University of Aeronautics and Astronautics.

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